SPICE line optical depths from MURaM

compute_spice_optical_depth.py calculates the line optical depth through each vertical MURaM column, including thermal broadening, optional microturbulence, the resolved velocity, and stimulated emission. It also synthesizes an optically thin comparison spectrum from the same atomic populations. The reusable physics and population-table code is in modules/spiceOpticalDepth.py.

This is an **equilibrium, bound-bound opacity diagnostic**. CHIANTI populations are not recomputed in response to the radiation trapped by an optically thick line. Consequently, a large optical depth flags a failure of the optically thin assumption; the saved comparison intensity is not a prediction of the emergent intensity in that regime.

Line selection and the instrument paper

The catalogue follows Table 1 of **SPICE Consortium et al. (2020), "The Solar Orbiter SPICE instrument: An extreme UV imaging spectrometer," A&A 642, A14**, DOI, accessible PDF. The paper specifies bands of 704–790 and 973–1049 Å. Its table is a selection of diagnostics, not every transition in those bands. Si XII is a second-order line: the absorbing photon has wavelength 520.67 Å, while its nominal first-order-equivalent detector coordinate is 1041.34 Å. All opacity and emissivity calculations use the physical photon wavelength.

CLI identifier	Ion	Paper wavelength (Å)	Paper log10 T (K)	Default
h_1_1025	H I	1025.72	4.0	
c_2_1036	C II	1036.34	4.3	
c_3_977	C III	977.03	4.5	yes
s_5_786	S V	786.47	5.2	yes
o_4_787	O IV	787.72	5.2	yes
o_5_760	O V	760.43	5.4	yes
o_6_1031	O VI	1031.93	5.5	yes
o_6_1037	O VI	1037.64	5.5	yes
ne_6_1005	Ne VI	1005.79	5.6	
si_7_1049	Si VII	1049.25	5.6	
ne_8_770	Ne VIII	770.42	5.8	yes
mg_8_772	Mg VIII	772.31	5.9	
mg_9_706	Mg IX	706.02	6.0	yes
fe_10_1028	Fe X	1028.04	6.0	
mg_11_997	Mg XI	997.44	6.2	
si_12_520	Si XII, order 2	520.67	6.3	
fe_18_974	Fe XVIII	974.84	6.9	
fe_20_721	Fe XX	721.55	7.0	

The eight defaults are my practical starting set for this repository's transition-region/coronal models. They include C III, the O VI doublet, and Ne VIII, plus intermediate-temperature diagnostics and Mg IX. This default set is a software choice. The table's approximate temperatures are labels, **not temperature masks or substitutes for an ionization balance**.

Use −−all−lines for all 18 entries or −−lines for a subset. H I is explicitly labelled an approximation when selected: Ly beta usually needs an NLTE hydrogen calculation, ideally using the simulation's hydrogen populations when available. C II and C III also need particular care with ionization equilibrium and radiative-transfer effects. The script does not automatically interpret the reader's H_1, HH, or HHp files, whose definitions must first be established for the particular simulation.

How this fits the existing repository

The implementation reuses MuramCube from modules/muram.py and follows the transpose used by MURaMData. It opens only the four required physical cubes. The full-domain input files are:

File	Quantity	Required unit
Header.<iteration>	Dimensions, spacings, simulation time	cm for spacings, s for time
eosT.<iteration>	Temperature	K
eosne.<iteration>	Electron density	cmˆ-3
result_prim_0.<iteration>	Mass density	g cmˆ-3
result_prim_1.<iteration>	Native vertical velocity vx	cm sˆ-1

An iteration is formatted with at least six digits, for example 400000. Iteration increments are not physical time intervals; the saved physical time comes from the header.

The low-level reader documents native axes (x, y, z) with **x vertical**. Applying .T gives (z, y, x): two horizontal axes followed by height, as in the other synthesis scripts. The line-of-sight cell length is therefore dX[0], including when the three header spacings differ. Positive vx points toward an observer at the upper x boundary, giving a blueshift. −−observer top implements this convention; −−observer bottom reverses the Doppler sign. Only vertical, axis-aligned rays are implemented. A slanted view requires resampling along rays and projecting all velocity components.

−−downsample strides the two horizontal dimensions. It does not average pixels and does not skip vertical cells or multiply their path lengths. Rectangular horizontal domains are supported.

The default integrates all vertical cells. −−height−start and −−height−stop select an explicit interval with Python's inclusive-start/exclusive-stop convention. For a solar observation, choose the lower boundary to exclude the subsurface convection zone according to your model's geometry. Index 150 in the existing scripts is not a universal photospheric height. There is no automatic temperature clipping: a cool foreground structure can absorb radiation from hotter material behind it, so dropping it changes the question being answered. The pre-existing MURaM tau field is not a SPICE line optical depth and is not used here.

Atomic populations and abundances

For element X and ion stage q, define

$$A_X = n_X/n_H, \qquad F_q(T) = n_{X,q}/n_X, \qquad p_i(T, n_e) = n_{X,q,i}/n_{X,q}.$$

The number density of hydrogen nuclei is obtained from mass density:

$$n_H = X_H \rho/m_H, \qquad n_i = n_H A_X F_q(T) p_i(T, n_e).$$

X_H defaults to 0.74 and is an explicit assumed composition parameter. Set −−hydrogen−mass−fraction to the value consistent with the model EOS. The script does not infer it from an abundance table. m_H = 1.00794 m_u is used; atomic weights for thermal widths and all other constants are declared in CGS in the module. No ion mass in kilograms enters a CGS thermal-width expression.

Using rho avoids a fixed n_H/n_e conversion, which would be particularly inappropriate in partially ionized plasma. n_H counts nuclei in all hydrogen ionization states, not only neutral hydrogen or protons.

For each ion, the code calls ChiantiPy.core.ion(..., temperature=T, eDensity=ne, abundance=...), uses IoneqOne for the installed default equilibrium balance, and calls populate() for the fractional level populations. Wgfa supplies transition wavelengths and Einstein A coefficients; Elvlc supplies level IDs and statistical weights mult = 2J+1. These interfaces are documented in the ChiantiPy source and the CHIANTI user guide, especially Sections 3.1 and 6.1–6.5.

The default abundance filename stem is sun_photospheric_2021_asplund, present in the locally installed CHIANTI 11.0.2 database. Choose a different installed file with −−abundance for a different composition or coronal FIP enhancement. Check the actual available filenames: older distributions can differ. The selected elemental abundance and equilibrium filename are written to metadata. populate() uses its installed default corrections and equilibrium proton density. Neither the MURaM electron-density cube nor a CHIANTI version number alone implies that non-equilibrium ionization or the advanced CHIANTI 11 ionization models are being used.

Transition identification

The default match window is **±0.15 Å around the paper wavelength**. Only positive (observed-energy) wavelengths and positive A coefficients are accepted. Theoretical wavelengths, stored as negative numbers in CHIANTI, are excluded. If no transition matches, the run fails rather than choosing an arbitrarily distant line. This interpretation of the files follows CHIANTI's radiative-data description.

Every matching same-ion component is retained and its opacity/emission is summed. This handles the closely spaced H I fine-structure components and the two selected Fe X components at 1028.053 and 1028.082 Å in the local data. The code prints and saves every selected wavelength, level pair, A coefficient, statistical weight, and effective oscillator strength. This is a feature selection window, not a full synthesis of every line falling anywhere within the final spectral window; other-ion blends are absent.

The local atomic-data check found C III 977.020 Å, O VI 1031.912 and 1037.613 Å, and Si VII 1049.140 Å. The latter differs from the paper by 0.110 Å and motivated the 0.15 Å matching default. In the local Si VII file it is the 2s2 2p4 3P1–2s2 2p4 1S0 transition (levels 2–5). All 18 catalogue features were checked against the installed database. These values are local-data findings, not changes to the paper's wavelengths.

The nearest selected component defines the exact reference wavelength and spectral grid center. Both this reference and the original paper wavelength are saved. For another database release, inspect the printed selections before increasing −−match−tolerance.

Population tables

The stored table is

$$Q_i(\log_{10} T, \log_{10} n_e) = A_X F_q(T) p_i(T, n_e) = n_i/n_H.$$

Only the levels needed by the selected transitions are retained. Lines of the same ion share a solve, including the O VI doublet. Default grids are log10(T)= 3..8 with 101 points and log10(ne)= 6..17 with 45 points. The interpolation is bilinear in the logarithms of T and ne, applied to the linear values of Q. It therefore preserves nonnegative corner values and handles exact zeros without taking their logarithms. Solves are skipped only where the tabulated equilibrium ion fraction is exactly zero.

The run rejects nonfinite/negative population values, solver errors, and populations that do not sum to unity. MURaM temperature, density, and electron density must be positive and finite. Velocities must be finite and below 0.01 c in magnitude. Model values outside the atomic table cause an explicit error; there is no extrapolation, edge clipping, or silent replacement by zero. Extend −−log−t/−−log−ne to cover your selected atmosphere. Bounds are checked while processing column blocks.

Population caches are compressed NPZ files named by a signature. The signature includes grid coordinates, selected transitions, ChiantiPy and database versions, numerical abundances, the equilibrium array, and hashes of files for the ion and next higher stage. −−rebuild−tables forces a rebuild. Cache writes use an atomic rename. A cache is a performance optimization, not an accuracy guarantee: compare maps with denser grids, particularly near steep ion-fraction gradients. Very faint ion-fraction tails can have large relative interpolation errors even when their absolute opacity contribution is negligible.

Opacity and spectral synthesis equations

For a transition from lower l to upper u, let λ_0 be the physical rest wavelength in cm and $\nu_0 = c/\lambda_0$. The Doppler width parameter is

$$b = \sqrt{2k_B T/m_X + \xi^2}, \qquad \Delta\nu_D = \nu_0 b/c.$$

Here ξ, set by −−microturbulence, is a **Doppler b parameter** in km/s, not a Gaussian standard deviation or FWHM. For this convention the Gaussian velocity standard deviation is $b/\sqrt{2}$. The temperature is used as both the electron temperature for populations and the ion temperature for broadening.

With $v_{\rm red} > 0$ away from the observer, the implemented center and profile are

$$\nu_c = \frac{\nu_0}{1 + v_{\rm red}/c}, \qquad \phi_\nu = \frac{1}{\sqrt{\pi}\Delta\nu_D} \exp\left[-\left(\frac{\nu - \nu_c}{\Delta\nu_D}\right)^2\right].$$

This centers the line at $\lambda_0(1 + v_{\rm red}/c)$, consistent with the nonrelativistic wavelength Doppler convention. Widths use the rest-frame ν_0; relativistic transfer factors are neglected. The profile is normalized over frequency, $\int \phi_\nu d\nu = 1$, and has units Hzˆ-1.

The extinction coefficient, including stimulated emission, is

$$\boxed{\alpha_\nu = \frac{\lambda_0^2 A_{ul}}{8\pi}\left(\frac{g_u}{g_l} n_l - n_u\right)\phi_\nu} \quad [\mathrm{cm}^{-1}].$$

Equivalently,

$$\alpha_\nu = \frac{\pi e^2}{m_e c} f_{lu}\left(n_l - \frac{g_l}{g_u} n_u\right)\phi_\nu, \qquad f_{lu} = \frac{m_e c}{8\pi^2 e^2}\frac{g_u}{g_l}\lambda_0^2 A_{ul}.$$

The implementation uses the A-based expression directly. The effective f derived by this relation is also recorded; this avoids relying on a gf field that can be absent or zero for a weak/forbidden transition. In particular, the lower-level occupation is explicitly solved: it is not assumed to equal the entire ion population for metastable transitions. Negative extinction from a population inversion is retained and flagged, rather than clipped away.

For cell-centered model data, the total column optical depth at each frequency is

$$\boxed{\tau(\nu) = \sum_{k=k_{\rm min}}^{k_{\rm max}-1} \alpha_{\nu,k}\,\Delta s_k}, \qquad \Delta s_k = \texttt{dX}[0].$$

Summing complete cells is intentional: a trapezoidal integral over cell centers would discard half of each boundary cell. A single-cell atmosphere still has optical depth $\alpha\Delta s$. The present file format supplies uniform spacings; a genuinely nonuniform grid needs a per-cell path-length implementation.

The spontaneous emissivity and optically thin comparison intensity are

$$j_\nu = \frac{h\nu_0}{4\pi} A_{ul} n_u \phi_\nu, \qquad I_{\lambda,\mathrm{thin}} = \frac{c}{\lambda^2} 10^{-8} \sum_k j_{\nu,k} \Delta s_k.$$

The factor 10^{-8} converts intensity per cm of wavelength to intensity per Å. **Optical depth itself gets no wavelength Jacobian**: tau2D simply samples τ_ν at $\nu = c/\lambda$. Both calculations include the actual column length and number densities, with no emission-measure normalization hidden in a contribution function.

The output does not solve $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$. In particular, multiplying the total thin intensity by exp(−tau2D) would incorrectly put all emission behind a foreground screen. Emission distributed along a ray requires depth-dependent transfer; resonance scattering additionally changes the source function and potentially the level populations.

Running the script

Use an environment with Python 3.9+, NumPy, SciPy, and ChiantiPy. Matplotlib is needed for `-plot` and the standalone visualization script. joblib is needed only for −−jobs other than 1. A CHIANTI atomic database is a separate installation from the Python package; see the official download instructions.

```
python −m pip install numpy scipy ChiantiPy joblib matplotlib
export XUVTOP=/path/to/CHIANTI
python compute_spice_optical_depth.py −−list−lines
```

The repository's .venv had none of these scientific packages during validation. An existing environment with them was available at /Users/memolnar/miniconda3/envs/SPICE_DEMs/bin/python, with the database at /Users/memolnar/Work/CHIANTI. No environments or database files were modified.

Start with a single C III snapshot and a sparse horizontal sample. The paths and iteration below follow example_CIII.py; adapt them to your machine:

```
python compute_spice_optical_depth.py \
  −−model−dir /glade/campaign/hao/radmhd/Rempel/Spot_Motion/AR_25Mm_v1000_vz300_disp/3
      D \
  −−iterations 400000 \
  −−lines c_3_977 \
  −−height−start 150 \
  −−downsample 8 \
  −−output−dir optical_depth_CIII \
  −−jobs 8
```

Here height index 150 is an example borrowed from the repo's existing boundary choice; establish that it is suitable for this snapshot before interpreting the column. −−height−stop defaults to the top of the model. Leaving out −−height−start includes the entire input column.

Run all eight default lines over an iteration range:

```
python compute_spice_optical_depth.py \
  −−model−dir /path/to/MURaM/3D \
  −−t−start 400000 −−t−end 502000 −−time−step 2000 \
  −−height−start 150 −−downsample 2 \
  −−output−dir optical_depth_default −−jobs 8
```

Or explicitly choose the resonance diagnostics:

```
python compute_spice_optical_depth.py \
  --model-dir /path/to/MURaM/3D --iterations 400000 402000 \
  --lines c_3_977 o_6_1031 o_6_1037 ne_8_770 \
  --height-start 150 --microturbulence 10 \
  --velocity-range 250 --num-waves 1001
```

To compute the whole catalogue, substitute --all-lines for --lines .... For a cooler/denser selected atmosphere or a convergence check, change the table grids, for example --log-t 3 8 201 --log-ne 5 18 105.

Build tables on their own, reusable in subsequent matching runs:

```
python compute_spice_optical_depth.py --tables-only \
  --lines c_3_977 o_6_1031 o_6_1037 ne_8_770 \
  --cache-dir opacity_tables --jobs 8
```

Table creation can be expensive for ions with many levels. Processes parallelize the density samples; snapshots and line features are processed sequentially. On HPC systems, constrain BLAS threads (for example OPENBLAS_NUM_THREADS=1 and OMP_NUM_THREADS=1) when using many worker processes, and allocate enough memory for each worker. --jobs -1 requests all available CPUs.

Input cubes stay memory mapped. --chunk-columns limits the atmospheric arrays held for integration, and wavelengths are streamed without allocating a full columns × heights × wavelengths profile array. Output arrays remain in memory for one line and snapshot: the two large float64 arrays together need about $16 * n_horizontal_1 * n_horizontal_2 * num_waves$ bytes, plus maps, table and compression overhead. A 512 × 512 field with 401 wavelengths needs about 1.57 GiB for those two arrays alone. Reduce horizontal sampling or wavelength count when necessary; chunking alone does not reduce output-array memory.

Existing output files are refused unless --overwrite is supplied. Different abundances, height cuts, or broadening choices should normally use different output directories so they can be compared. Files required for every selected iteration are checked before building tables.

Outputs and interpretation

One file is saved per feature and snapshot: spice_opacity_c_3_977_400000.npz, for example.

Key	Shape	Meaning
tau2D	`(1, native_z, native_y, n_wavelengths)`	Total selected-column spectral optical depth
I2D	same	Optically thin intensity, erg sˆ-1 cmˆ-2 srˆ-1 Åˆ-1
tau_rest	(1, native_z, native_y)	Optical depth at the exact atomic reference wavelength, including model velocities
tau_peak	same map shape	Maximum optical depth among the sampled wavelengths
tau_static	same map shape	Optical depth at the reference wavelength with all resolved velocities set to zero
wvlGrid	(n_wavelengths,)	Increasing physical wavelengths in Å
velocityGrid	(n_wavelengths,)	Offsets relative to the atomic reference in km/s, red positive
detectorWvlGrid	(n_wavelengths,)	order ∗ wvlGrid, a coordinate only, not an instrument response

Key	Shape	Meaning
metadata	scalar JSON string	Atomic selections, units, parameters, versions, file hashes, geometry, diagnostics and command

The leading dimension is length one for compatibility with the repository's time-leading I2D convention. Horizontal dimensions include any downsampling. These files can supply I2D and wvlGrid to the existing visualization script, but their intensity model must be kept in mind. The metadata loads with allow_pickle= False; no Python object deserialization is required.

tau_static is a useful comparison when velocity gradients move absorbing layers out of resonance. It includes thermal and selected microturbulent broadening. For a multiplet it is the sum of component opacities at the **shared reference wavelength**, not a sum of each component's individual peak.

tau_peak depends on the chosen finite spectral grid and is not a guaranteed continuous maximum. The script warns when resolved velocities exceed the spectral window or when the sample spacing is greater than one third of the smallest Doppler b. These are basic checks: extend the window enough to capture the wings and repeat with finer sampling until the maps and integrated intensities converge. Even in the absence of a warning, a near-edge line can lose wing flux. The odd default of 401 samples guarantees the exact reference wavelength is present.

Matplotlib maps of maximum column optical depth

Add `-plot` to a synthesis command to save a PNG beside every NPZ:

```
python compute_spice_optical_depth.py \
  --model-dir /path/to/MURaM/3D --iterations 400000 \
  --lines c_3_977 --height-start 150 --downsample 8 --plot
```

The image is named `spice_opacity_c_3_977_400000_tau_peak.png`. To visualize already saved results without repeating synthesis:

```
python visualize_spice_optical_depth.py optical_depth_output/spice_opacity_*.npz \
  --output-dir optical_depth_maps
```

The standalone command reads only `tau_peak` and `metadata`, and requires NumPy and Matplotlib. It does not load the spectral cubes, CHIANTI, or MURaM inputs. Use `-overwrite` to replace existing images. Both commands support `-plot-scale log|linear`, `-plot-limits MIN MAX`, `-plot-cmap`, and `-plot-dpi`. Defaults are logarithmic grayscale, fixed optical-depth limits of `1e-3` to `5`, and 180 DPI. Values at or below `1e-3` are black and values at or above `5` are white; the colorbar spans those same limits and has the map's height. Limits are optical depths, not their logarithms. Use `-plot-limits` or `-plot-cmap` to choose a different display scale.

Each pixel shows

$$\tau_{\mathrm{peak}}(x, y) = \max_{\lambda} \int_{\mathrm{LOS}} \alpha_{\lambda}\, ds.$$

For nonnegative extinction, cumulative optical depth is largest through the complete selected column. The map then takes its maximum over the sampled wavelengths; it is not a maximum of the local extinction coefficient. It does not calculate a depthwise maximum of cumulative optical depth for inverted populations, where extinction can be negative.

The plot labels the two horizontal coordinates **x [Mm]** and **y [Mm]**: plot x is native MURaM y, plot y is native MURaM z, and native MURaM x remains the line of sight. Stored maps have row/column order `(native_z, native_y)`, so they are plotted without transposition. Coordinates are relative to the domain's horizontal

lower boundaries. Sampled cell centers are calculated as `(native_index + 0.5) * cell_spacing_cm / 1e8`; display edges lie halfway between sampled columns and end at the domain boundaries. Unequal spacings, rectangular domains, and downsampling factors that do not divide the domain size are supported. Colors represent nearest sampled columns, not pixel averages.

A white contour marks $\tau_{\mathrm{peak}} = 1$ when the finite map spans that value and both horizontal dimensions have multiple samples. In logarithmic maps, finite values at or below the lower limit, including zero and negative values, are black; nonfinite pixels are gray and their count is annotated. Linear colors retain finite negative optical depths. Rendering uses a noninteractive Matplotlib canvas, so synthesis on an HPC compute node does not open windows.

Read a result and use the plotting helper from Python:

```
import json
import numpy as np
from scipy.integrate import trapezoid
from visualize_spice_optical_depth import save_tau_peak_map

with np.load("optical_depth_CIII/spice_opacity_c_3_977_400000.npz",
             allow_pickle=False) as data:
    tau_peak = data["tau_peak"][0]
    tau_static = data["tau_static"][0]
    integrated_thin = trapezoid(data["I2D"][0], data["wvlGrid"], axis=-1)
    metadata = json.loads(str(data["metadata"]))

print("Fraction of sampled columns with tau_peak >= 1:", (tau_peak >= 1).mean())
print("Atomic components:", metadata["atomic"]["components"])
save_tau_peak_map(tau_peak, metadata, "CIII_tau_peak.png", limits=(0.01, 100))
```

For non-inverted populations, tau << 1 means weak line extinction along the selected column, while tau >= 1 indicates potentially consequential opacity. The total optical depth through a column can exceed the optical depth between a particular emitting layer and the observer. These products do not locate a tau-one surface or save cumulative depth-dependent optical depths.

The O VI doublet provides a useful internal comparison. With negligible upper populations, equal lower-level columns, and the same velocity broadening, its static peak-depth ratio scales as f $*$ wavelength, approximately two. It is not an assumption enforced by the code, and observed intensity ratios also depend on excitation and transfer.

Limits and validation

The calculation assumes a Maxwellian electron distribution, equilibrium ion fractions, statistical-equilibrium level populations, a single ion/electron temperature, and Gaussian profiles. It includes neither an external pumping radiation field nor a self-consistent scattering source function. Natural and pressure wings, partial redistribution, non-equilibrium ionization, and instrumental PSFs/effective areas are absent. It is not a calculation of SPICE detector counts. Absence of line opacity does not by itself establish that an observed spectrum is described fully by collisional emission.

Continuum opacity is also absent. In particular, neutral-hydrogen bound-free absorption can affect the short-wavelength band below the Lyman edge near 912 Å and the physical Si XII wavelength. A full foreground absorption model would need the appropriate neutral populations and continuum cross sections. The saved optical depths refer exclusively to the selected bound-bound components.

The test suite uses analytic slabs and artificial MURaM-format files. It checks absolute line-center normalization, total cell length (including a one-cell column), stimulated emission and inversions, thermal and microturbulent widths, Doppler shifts, multiplet addition, profile normalization, and the wavelength emissivity

Jacobian. A nonsquare fixture with unequal grid spacings checks the reader transpose, vertical path length, height slicing, and observer sign. Invalid cells, out-of-grid values, missing inputs, and output collisions are also exercised.

python −m unittest discover −s tests −v
SINSPICE_TEST_CHIANTI=1 python −m unittest discover −s tests −v

The second command additionally checks actual catalogue identifications and builds a small C III table, compares a tabulated population against a direct ChiantiPy solve, verifies cache reuse, and runs the CLI workflow on a generated MURaM-format snapshot, including its PNG output. Plotting tests check the map orientation and physical extent with unequal spacings and non-divisible downsampling, zero/negative/nonfinite pixels, and standalone plotting from an NPZ containing only the map and metadata. All 15 tests passed using ChiantiPy 0.15.2 and CHIANTI 11.0.2. Separate smoke runs built tables for all 18 features on a 2×2 grid and built the full default 101×45 C III table with two worker processes. The tiny smoke-test grids are deliberately insufficient for science; they exercise the interfaces and solver paths. Artificial fixtures are not solar-model validation. Production MURaM snapshots at the hard-coded GLADE paths were not available locally, so no physical opacity maps for those simulations are claimed here.

For a scientific production run, first establish the correct geometry and composition, then converge the population grids and spectral sampling. Use non-LTE/NEI calculations where the inferred optical depths or timescales demand them. Record the saved atomic metadata alongside any reported results.